2 Week 2: Formal & Informal Technical Reports

2.1 The Executive Summary and Abstract.

An abstract is a short and complete summary of a research paper or IT project. Even though it is the last part you write, it is the first thing that professors, reviewers, and academic search engines read Dupree and Casapao (2023), Li (2015). A good abstract must be written clearly and include four main parts: the problem you are solving, the technology or method you used, your actual results or test data, and your final conclusion. It should be independent and easy to understand so readers can quickly see what your project achieved without reading the whole report.
In academic research and industrial engineering documentation, abstracts are split into two primary types:

1. **The Informative (Result-Driven) Abstract**: An informative abstract acts as a complete, miniature version of a research paper. It summarizes everything in a short space: **the goal, methods, who or what was tested, the actual results, and the final conclusion**.
   It is designed so readers can understand exactly what you discovered without needing to open the full report. These are often organized with clear subheadings: Background, Methods, Results, and Conclusions Haynes et al. (1996).
   Why use an informative abstract?

   . It helps professors and graders quickly see your project's design and results.

   . It helps other researchers immediately spot if your study is relevant to them.

   . It makes it much easier for academic search engines to find your paper because it contains specific details and metrics Haynes et al. (1996).

   Example: Imagine you are writing a graduation project analyzing how 1,200 different companies manage their digital files and data (Enterprise Content Management).
   An informative abstract for this project would explicitly state:

   (a) **Purpose:** Why managing digital files matters to businesses.

   (b) **Method:** How you sorted and looked through the 1,200 case reports.

   (c) **Key Findings:** What differences or trends you found in the technology used across different industries.

   (d) **Conclusion:** What these gaps between theory and real-world practice mean for the future of IT Jaakonm"aki et al. (2018).

The Informative (Result-Driven) Abstract	Corresponding IMRD Chapter	What it Answers
Background & Aim	**I**ntroduction	*Why did you do this project?.*
Method / Scope	**M**ethods	*What did you build or test?.*
Results	**R**esults	*What did your metrics show?.*
Conclusion	**D**iscussion / Conclusion	*What do these findings mean?.*

Table 1: Result-Driven Abstract

2. **The Descriptive (Indicative) Abstract**: A descriptive abstract, on the other hand, is just an overview. It tells the reader what the paper is about, but it **stops short of giving the actual results or data**. It introduces the general topic, scope, or research questions, but doesn't tell the reader the final answer. Because it leaves out the data, descriptive abstracts are usually much shorter.

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2.2 Disclaimer, Acknowledgment, Financial Support

1. **Writing a Disclaimer** A disclaimer is used to clarify limitations, responsibilities, or disclaim liability concerning the content or use of the research. It is often brief and formal, stating that the views expressed are those of the authors and do not necessarily reflect those of affiliated institutions or funding bodies.

   How to Write:

   * Clearly state limitations or non-endorsement of views.
   * Can mention responsibility disclaimers around data, opinions, or errors.
   * Usually placed after the abstract or acknowledgments section.

   Example: "The views expressed in this thesis are those of the author and do not necessarily reflect the official policy or position of the affiliated institution or funding agencies. The author assumes full responsibility for the content of this work." This type of statement acknowledges the scope and limitations of the research while protecting institutions or sponsors from unintended liability or misinterpretation.

2. **Writing an Acknowledgment Section**: The acknowledgment section recognizes people, organizations, or institutions that provided academic, moral, technical, or financial support during the research process. This section is essential in scholarly writing to show gratitude and disclose collaborative contributions.
   Key Points in Scientific Acknowledgments:

   * Thank advisors, mentors, and supervisors for academic guidance.
   * Mention peers or colleagues who contributed feedback or assistance.
   * Include institutional resources or libraries if applicable.
   * Recognize funding bodies or financial supporters explicitly but separately (often in a financial support section).
   * Maintain a formal and sincere tone.

   Content Structure:

   * Start with general thanks to advisors or committee.
   * Recognize colleagues and technical or moral supporters.
   * Include institutions and library resources.
   * Optionally, mention family or personal support if culturally appropriate.

   Example (from ETDs research demonstrating forms of support acknowledged): "I am deeply grateful to my advisor, Prof. X, for insightful guidance and unwavering support. My thanks also extend to the research group members for their constructive feedback and encouragement. The library staff's assistance with accessing key resources was invaluable. Finally, I appreciate the moral and emotional support from family and friends throughout this journey."Lamba and Erz (2025).

   Patterns Across Disciplines:

   * Acknowledgments recognize academic, moral, financial, and sometimes religious support.
   * Gender-based studies found differences in emphasis on academic vs. moral support Lamba and Erz (2025).

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* Technical assistance is often acknowledged separately, especially in experimental sciences Díaz-Faes and Bordons (2014).

3. **Reporting Financial Support / Funding Statements**: Explicitly acknowledging financial support or funding sources is critical to maintain transparency and uphold ethical standards in research.
How to Write:

* Name the funding agency/organization explicitly.
* Include grant numbers or project identifiers if applicable.
* State the extent or role of the funding (if relevant).
* Sometimes include disclaimers about the funders' role in design or decision-making.

Common Phraseology:
"This research was supported by [Funding Agency], grant number XXXX. The funding body had no role in the design of the study, collection, analysis, or interpretation of data."
or
"We gratefully acknowledge financial support from the [Name of Institution/Foundation] for making this study possible."
Examples from literature:

. Systematic reviews show common reporting of government, philanthropic, and academic institution funding sources in scientific papers Zanca et al. (2005).
. Explicit funding acknowledgments improve transparency and are often required by journals or academic regulations Díaz-Faes and Bordons (2014), Walsh and Sanchez (2009).

Additional Notes:

. Some databases, like Web of Science, index funding acknowledgments separately, highlighting their importance in research metadata Hubbard and Laddusaw (2020).
. The phrasing should be formal and concise.

2.3 Technical Keyword Extraction, System Data Filtering, Academic Brevity Benchmarks

1. **Technical Keyword Extraction**: **What it means:** Automatically finding the most important words or phrases that sum up the core topic of your paper. Think of them as the tag words that help search engines like Google Scholar index your project so other developers can find it.
How it works:
Keyword Training: isolate high-value technical terms by systematically stripping away linguistic noise and identifying core system components.
Examples:

High Quailty : "This project deployed an automated cloud migration script using Terraform to move an enterprise database to AWS hosting, reducing core server latency by 22%."

Junk : "We built an incredibly amazing, super-fast, and really beautiful database system that works perfectly well and handles our text data wonderfully."

1. The Clutter Filter (Supervised Action): Instantly drop descriptive fluff (adjectives/adverbs like amazing, wonderful, fast). Keep only concrete engineering nouns.
2. The Frequency Scan (Unsupervised Action): Look for repeating core terms (like cloud network or database cluster). If it appears multiple times, it carries the heaviest weight.

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3. The Context Check (Deep Learning Action): Look at surrounding words to lock down the exact technical meaning (knowing if port means a software channel or a shipping dock by looking at words like firewall).

2. **System Data Filtering**: **What it means:** Cleaning raw data streams to remove background noise, junk logs, or useless information so your software application only processes clean, highly relevant data.
How it works:

   (a) **Dictionary-Based (Ontology) Filtering:** Using a predefined map of technical concepts to make sure your system understands word meanings and doesn't get confused by words with double meanings.

   (b) **Signal Clean-up (Statistical Smoothing):** Using mathematical formulas to smooth out jumpy raw data coming from hardware sensors.

   (c) **User Recommendations:** Systems that predict what a user wants by analyzing their profile history or matching them with similar users (Collaborative Filtering).

   Project Example: Imagine you are building an IoT weather monitoring station using an Arduino and a temperature sensor. Out in the wind, the sensor fluctuates erratically due to electrical noise, reading: 22°C, 85°C, 22°C, -10°C. Applying a statistical filter (like a smoothing filter) cleans this stream down to a realistic, accurate line: 22.1°C, 22.2°C, 22.2°C. If you handle massive live streams, you use high-speed tools like Apache Kafka to drop this data into your app without lagging.

3. **Academic Brevity Benchmarks**: **What it means:** Setting clear, measurable goals to keep your technical writing short, compact, and packed with facts, instead of filling pages with fluff and unnecessary words.
How to do it in your Thesis:
• Cut out the Fluff: Watch your use of descriptive words (adjectives and adverbs). Computer science reports don't need decorative writing.
• Count Your Words: Look at successful past graduation theses from your department to see their average lengths per section.
IT Project Example:
• **Fluffy & Cluttered (Bad):** "We built a truly amazing, incredibly fast, and very beautiful cloud architecture database system that handles information wonderfully well."
• **Brief & Dense (Good):** "A multi-tier cloud database architecture was deployed, achieving a 22% reduction in query processing latency."

2.4 The Grammar of Abstract.

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Abstract Move (IMRD)	Recommended Tense	Underlying English Grammar Structure	Student-Facing Example
Background & Aim	Present Simple	**Subject + Verb 1** (s/es) *(Used for active, ongoing industry facts and project goals.)*	"Cloud computing [Subject] **presents** [Verb 1] security challenges. This project [Subject] **proposes** [Verb 1] a hybrid architecture..."
Method / Scope	Past Simple (Passive)	**Subject + was/were + Verb 3** (Past Participle) *(Used to show what was done to the technical system.)*	"A pilot environment [Subject] **was built** [was + Verb 3] using AWS instances, and configurations [Subject] **were deployed** [were + Verb 3]."
Results	Past Simple (Active)	**Subject + Verb 2** (Past Tense form) *(Used to show what the metrics or system did during testing.)*	"The deployment [Subject] **reduced** [Verb 2] total server overhead by 22% and **stabilized** [Verb 2] memory consumption."
Conclusion	Present Simple	**Subject + Verb 1** (s/es) *(Used to explain what your final data means right now.)*	"These findings [Subject] **indicate** [Verb 1] that optimization **is** viable for large enterprise migrations."

Table 2: Split-Tense Protocol Matrix

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